

1/8/26 Course: Computer Vision
Course code: ITA0506

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CLASS
TEST

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Slot: D

① Image Enhancement: Image enhancement in the spatial domain is the process of improving the visual quality of an image by directly modifying the pixel values.

Spatial domain Formula:

Where:

$f(x,y)$ = Input Image

$g(x,y)$ = Enhanced Image

T = Transformation function.



Techniques

a) Grey-Scale Transformations

Changes the brightness and contrast of an image.

Ex: Image Negative, Log Transformation

Applications: Medical imaging, Satellite images

Advantages: Simple implementation, Improves brightness & contrast

b) Histogram Processing

Histogram shows the distribution of pixel intensities.

Techniques: Histogram Equalization, Histogram Matching

Applications: Low contrast images, CCTV images, Medical images

Advantages: Improves global contrast, Enhances visibility

c) Spatial Filtering:

Uses a filter mask to modify neighbouring pixels.

Types: Smoothing and Sharpening Filters

Applications: Noise removal, Edge Enhancement

Advantages: Improve image quality

Stability: Grey-level : Brightness improvement

Conclusion: Spatial domain enhancement directly processes image pixels to improve image quality. Grey-level Transformation adjust brightness, histogram processing

Improves contrast and spatial filtering removes noise.

② Image Smoothing:

Image Smoothing removes noise and small variations from an image.

Working principle:

- Uses neighbouring pixels
- Computes average or weighted average
- Produces a blurred image

Filters:

- Mean Filter
- Gaussian filter
- Median Filter

Advantages:

- Remove noise
- Smooth appearance

Applications

- Medical imaging
- Satellite images
- Preprocessing

Image Sharpening: Highlights edges and fine details

Working principle: Uses a derivatives to emphasize rapid intensity change.

Filters: Laplacian, Sobel, Prewitt Filters

Advantages: Highlights edges,

Applications: Improves fine details

↳ Face Recognition, OCR, object detection.

Difference:

Image Smoothing
Removes noise
produces blur
Uses averaging

Image Sharpening
Enhances edges
Sharpens image
Uses derivatives

Conclusion: Smoothing is used before analysis to reduce noise, while Sharpening is used to improve edge details.

Frequency Domain Enhancement: It processes an image after converting it into the frequency domain using the Fourier Transform.

Process: Image $\rightarrow$ Fourier Transform $\rightarrow$ Filtering $\rightarrow$ Inverse Fourier Transform

Low-pass Filter (LPF)

Allows low frequency components and blocks high-frequency components

Effect: Removes noise, produces blur

Applications: Noise Reduction, Medical imaging

Advantages: Smooth images, Removes random noise

High-pass Filter (HPF)

Allows high frequency and suppresses low frequencies

Effect: Highlights edges, Improve details

Applications: Face Recognition, Industrial inspection

Advantages: Better edge visibility.

Comparison:

LPF

- ① Removes noise
- ② Produces smooth image
- ③ Remove high frequency

HPF

- ① Enhances edges
- ② produces sharp image
- ③ Remove low frequencies

Conclusion: LPF is mainly used for smoothing, while HPF is used for edge enhancement.

④ Image Segmentation: It divides an image into meaningful regions for easier analysis

Techniques:

(i) Point Detection: Detect isolated pixels

Application: Dead pixel detection

(ii) Line Detection: Detects horizontal, vertical and diagonal lines

Application: Road and Lane detection

(iii) Edge Detection: Detect object boundaries

Methods:

• Sobel

• Prewitt

• Canny

Applications: object recognition, medical images

(iv) Thresholding: Separates foreground from background using intensity values.

Applications: OCR, Document processing

(v) Region-Based Segmentation:

Groups neighboring pixels having similar properties

Applications: Tumor detection, Satellite images

Comparison:

Technique	Advantage	Limitation
Point Detection	Simple	Detects only isolated points
Line Detection	Detect lines	Sensitive to noise
Edge Detection	Good boundaries	Noise sensitive
Thresholding	Fast	Fails with uneven light
Region-Based	Accurate segmentation	Computationally expensive

Best Technique: Edge detection

Conclusion: Image segmentation simplifies image analysis by separating objects from the background using different techniques.

Edge Detection: Detects sudden changes in pixel intensity

Methods:

• Sobel , prewitt , Canny

Applications: Face Recognition, Traffic Sign detection

Edge linking: Connects broken edge pixels to form continuous edges.

Methods: Hough Transform, Morphological Operations

Applications: Object Tracking, Shape detection

Boundary Detection: Extracts the complete out line of the object.

Applications:

- Tumor boundary detection
- Industrial inspection
- Object measurement

Importance:

- Identifies object boundaries
- Improves object recognition
- Supports defect detection

Example: Traffic Sign Detection

Original image



Edge Detection (find sign edges)



Edge linking (connect broken images)



Boundary Detection (extract complete sign)



Object Recognition (identify traffic sign)

Conclusion:

Edge detection identifies the intensity changes, edge linking connects fragmented edges, boundary detection